



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
FULL SYLLABUS
08-07-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE : [SCORE-II (ALL STAR BATCH)]

PAPER-1

Time : 3 Hours

Maximum Marks : 186

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 12)

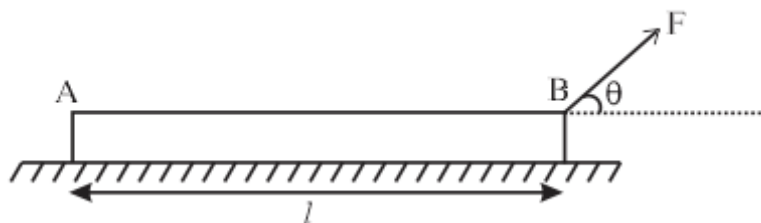
- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. A force F is applied on a uniform horizontal rod of mass m kept on horizontal rough surface as shown. Coefficient of friction varies as $\mu_0 x$, where x is distance from end A of the rod.

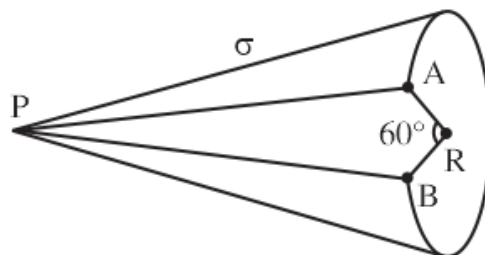


What is the minimum value of force so that rod starts moving ? (Rod does not topple)

- (A) $\frac{\mu_0 mg l}{\sqrt{1 + \mu_0^2 l^2}}$ (B) $\frac{mg}{\sqrt{1 + \mu_0^2 l^2}}$ (C) $\frac{\mu_0 mg l}{2\sqrt{1 + \mu_0^2 l^2}}$ (D) $\frac{mg}{4}$

Space for Rough Work

2. A non-conducting hollow cone has uniform surface charge density σ . A part ABP is cut and removed from the cone. The potential due to remaining portion of the cone at point 'P' is



- (A) $\frac{5\sigma R}{6\epsilon_0}$ (B) $\frac{5\sigma R}{24\epsilon_0}$ (C) $\frac{5\sigma R}{3\epsilon_0}$ (D) $\frac{5\sigma R}{12\epsilon_0}$
3. Expression of Maxwell's velocity distribution for number of molecules per unit velocity is given as $\frac{dN}{dv} = kv^2 \exp[-bv^2]$ for an ideal gas. Here, k and b are constants for a gas at given temperature. What will the value of $b \times \frac{RT}{M}$? (R is universal gas constant)
- (A) 0.5 (B) 2.55 (C) 0.39 (D) 2
4. A radioactive sample is undergoing α decay. At any time t_1 its activity is A and at later instant t_2 its activity is $\frac{2A}{3}$. Which of the following is **INCORRECT** ?
- (A) Mean life of sample is $\frac{t_2 - t_1}{\ln(\frac{3}{2})}$
- (B) Half life of sample is $(t_1 + t_2) \frac{\ln 2}{\ln(\frac{3}{2})}$
- (C) Probability of survival of a nucleus after one mean life is $\frac{1}{e}$
- (D) Activity will become half of initial activity after one half life

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

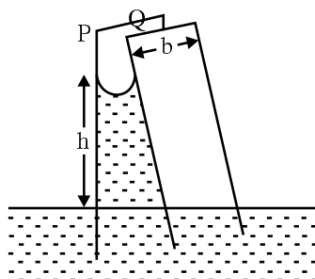
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

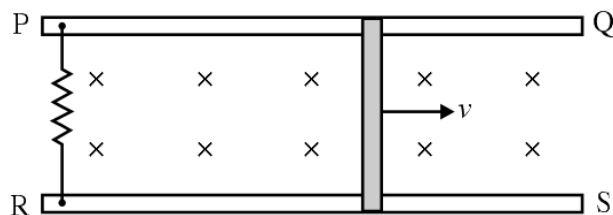
5. Two plates P and Q are placed in a liquid as shown in figure. Plate P is kept vertical and plate Q makes angle θ with vertical. The plates are placed very close to each other.



Due to surface tension 'S' of liquid, the liquid rises to a height h above the water level in vessel. A horizontal line between the plates drawn at point of contact of liquid on plate P measures d . Angle of contact of the liquid is zero and density of liquid is ρ . Take atmospheric pressure as P_0 and breadth of plate $b \gg d$. Which of the following is/are INCORRECT?

- (A) The height $h = \frac{2S}{\rho g}$
- (B) The height $h = \frac{S(1 + \cos \theta)}{\rho g}$
- (C) The horizontal force exerted by liquid on plate Q is $\left(P_0 + \rho g \frac{h}{2}\right) hb$
- (D) The vertical force exerted by plates on plate Q is $\left(P_0 + \rho g \frac{h}{2}\right) hb \tan \theta$

6. In the shown figure PQ and RS are two parallel conducting rails 1m distance apart placed in horizontal plane. A uniform magnetic field of strength 1T is present perpendicular to the plane of rails. The ends of the rails are connected via a resistor of resistance $R = 1\Omega$. A conducting rod of length 1 m and mass 1kg is placed on the rails. The coefficient of friction between the rails and rod is 0.2.

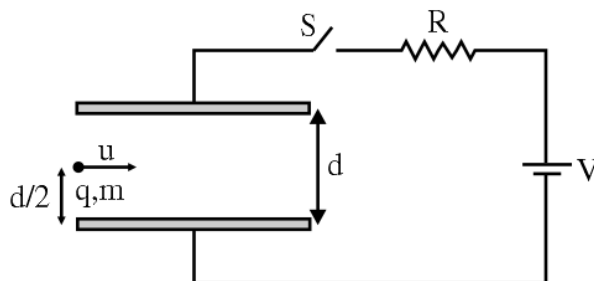


The rod is now given a velocity of 2 m/s perpendicular to rod over the rails towards right.

- (A) The distance travelled before stopping is 0.61 m
- (B) The distance travelled before stopping is 1.39 m
- (C) The time taken to stop is $\ln 2$ seconds
- (D) Heat lost in resistance is 0.77 J

Space for Rough Work

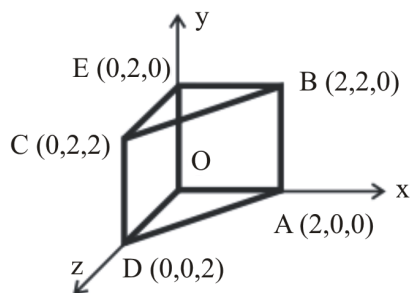
7. In the figure shown, a particle of charge 'q' and mass 'm' enters with velocity 'u' in an uncharged capacitor of capacitance 'C' midway between the plates. Its initial velocity is parallel to plates of capacitor. The switch is closed at $t = 0$. It is known that time spent by particle between the plates of capacitor (t_0), before it strikes one of the surface, is very small in comparison to RC (Time-constant of circuit). Neglecting gravity and any change in charge distribution on plates due to q, which of the following is/are correct?



- (A) Horizontal distance travelled by charge just before it hits the plate is $u \left(\frac{3md^2RC}{qV} \right)^{1/3}$
- (B) Acceleration of particle at any time 't' is given by $\frac{qVt}{mdRC}$
- (C) Voltage across capacitor at any $t < t_0$ is $\frac{Vt}{RC}$
- (D) Velocity of particle at any time 't' is $\frac{qVt^2}{2mdRC}$

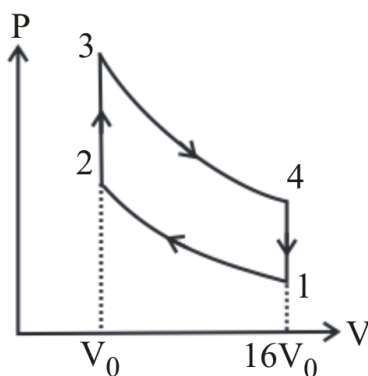
Space for Rough Work

8. A right angle prism is kept in an electric field $\vec{E} = x\hat{i} + y\hat{j} - z\hat{k}$ N/C as shown. All dimensions in figure are in meter ?



Which of the following is/are true ?

- (A) Flux through surface CEB is 4 Vm (B) Flux through surface ABCD is 4 Vm
(C) Flux through surface ABCD is 8 Vm (D) Charge enclosed by prism is $4\epsilon_0$
9. Equal moles of Helium and Oxygen are made to undergo through a cyclic process as shown in P-V graph. this graph represents an ideal Otto cycle on which petrol engines work. Process 1-2 and 3-4 are adiabatic and process 2-3 and 4-1 are isochoric. What is the value of efficiency of the cycle ?



- (A) $1 - \frac{1}{2^{4/3}}$ (B) $\left(1 - \frac{1}{\sqrt{2}}\right)$ (C) $1 - \frac{1}{2\sqrt{2}}$ (D) $\frac{3}{4}$

Space for Rough Work

10. A 6 cm long illuminated object AB of negligible thickness is placed on the principal axis of a converging lens of focal length 40 cm. Diameter of lens of circular aperture is 4 mm. End A of the object is at 50 cm from lens. Image of object is 10 times its length. Which of the following is/are correct ? (In relevant options, when screen is placed it is perpendicular to principal axis)
- (A) End B is at a distance of 44 cm from lens
- (B) End B is at a distance of 56 cm from lens
- (C) If screen is placed 140 cm from lens, image on it will be a circular disc of diameter 1.2 mm
- (D) Screen should be placed at 200 cm from lens so that size of image (disc) on screen is minimum
11. A pressure difference P is needed across both ends of a pipe to make liquid flow in a tube of length ℓ and inner radius r with constant speed. According to Poiseuille's equation volume flow rate (v) in such a case depends on pressure difference (P), length of tube (ℓ), inner radius of the tube (r) and coefficient of viscosity (η). Which of the following can be possible correct representation of relation ? (here k is dimensionless constant)
- (A) $v = k \frac{\eta r^2 P}{\ell}$ (B) $v = k \frac{P r^3}{\eta \ell}$ (C) $v = k \frac{P r^4}{\eta \ell}$ (D) $v = k \frac{\eta r^3 P}{\ell}$
12. In a moving coil galvanometer, magnetic field of permanent magnet is B , number of loop in coil is N , Area of loop is A and torsional constant of spring in it is C . Which of the following are correct?
- (A) If B and N are doubled and A and C are halved then current sensitivity will becomes 4 times.
- (B) If B and A are doubled and N and C are halved then current sensitivity will becomes 4 times.
- (C) If B and C are doubled and N is halved then voltage sensitivity will remain same.
- (D) If B is doubled and N and C are halved then voltage sensitivity will remain same.

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

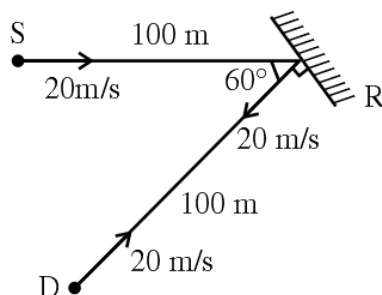
Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

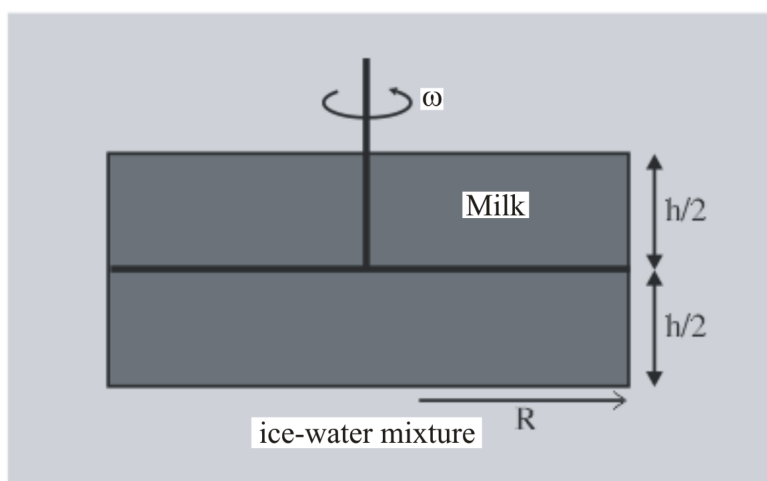
-
1. The potential energy U in joule of a particle of mass 1 kg moving in xy plane obeys the law $U = 3x + 4y$ where x, y are coordinates of the particle in meter. If the particle is released from rest at $(6, 4)$ its trajectory intersects x -axis at $(\alpha, 0)$ metres and y -axis at $(0, \beta)$ metres. Find $|\alpha| + |\beta|$.
 2. A liquid is filled in a bottle to the brim and the cap is placed tightly. The bottle has a linear thermal expansion coefficient of $10 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$ and coefficient of volume expansion of liquid is $630 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$. On increasing the temperature of bottle and liquid inside by 20°C , the magnitude of normal force in newton per square centimeter is $\beta \times 10^3$. [Bulk modulus of liquid is $2 \times 10^9 \text{ N/m}^2$]. Find β
-

Space for Rough Work

3. A source S, a detector D and a reflector R are moving as shown in figure. If source emits a frequency $f_0 = 264$ Hz then what will be beat frequency (in Hz) detected by detector D at this given instant ? (Speed of sound is 340 m/s)

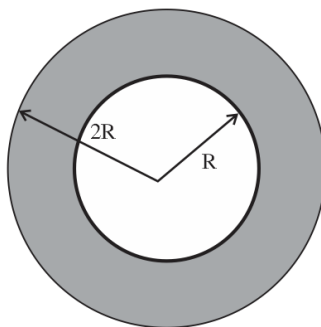


4. To prepare icecream, milk of density ρ is being churned by rotating a disc of radius R and negligible thickness with constant angular velocity. All the heat generated in doing work against viscous force is converted into heat. Cylindrical container in which milk is kept has radius R and height h. Surface of container has thickness t_0 and has thermal conductivity K. Initially temperature of milk is T_0 and whole system is placed in ice water mixture. Calculate the time (in s) after which temperature of milk drops by 20°C ($K = 0.4 \text{ W/}^\circ\text{C}\times\text{m}$, $T_0 = 50^\circ\text{C}$, $h = 4 \text{ mm}$, $\eta = 10^{-1}$ poise, $t_0 = 1 \text{ mm}$, $\rho = 10^3 \text{ kg/m}^3$ specific heat of milk $= 4 \times 10^3 \text{ J/kg} \times ^\circ\text{C}$, $\omega = 100 \text{ rad/s}$, $\ln 2 = 0.693$, $R = 40 \text{ cm}$) (Assume that density and viscosity of milk does not change with temperature).



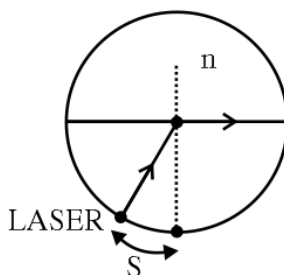
Space for Rough Work

5. A charged conducting sphere of radius R is surrounded by a spherically symmetric region of dielectric of radius $2R$. The region of dielectric has 20% of the total energy stored.



Find the dielectric constant of the material.

6. A horizontal cylindrical tank $\frac{6}{\pi}$ m in diameter is half filled of water. The space above water is filled with pressurised gas of unknown refractive index. A small laser can move on the curved bottom of the water and aims a light beam towards the centre of the water surface. When the laser has moved a distance $s = 1$ m or more (measured along curved surface) from lowest point in the water, no light enters the gas. If the refractive index of gas is $\frac{2}{\sqrt{n}}$. Find value of n ($\mu_{\text{water}} = \frac{4}{3}$)



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 12)

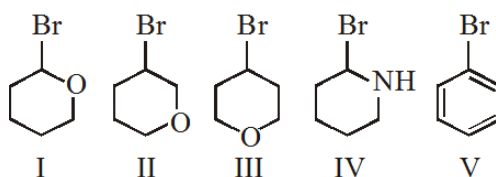
- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

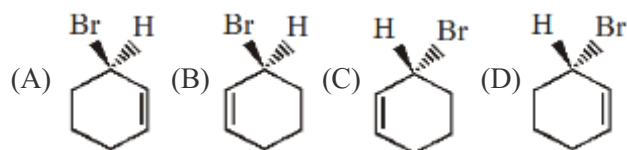
Negative Marks : -1 In all other cases

1. Ease of SN^1 reactions among given compounds upon treatment with aq. NaOH will be in the order as



- (A) $IV > I > III > II > V$ (B) $I > II > III > IV > V$
(C) $I > IV > III > II > V$ (D) $V > IV > III > II > I$

2. Which of the following statement about the relationships of the structure (A), (B), (C) and (D) (given below) is **incorrect** ?



- (A) A and C are enantiomers
(B) A and C are identical
(C) B and D are identical
(D) A and B are enantiomers

Space for Rough Work

3. **Statement-1** : In leaching of red bauxite KOH or HNO_3 can be used as a leaching agent.
Statement-2 : Al_2O_3 is amphoteric, it reacts with acid and base both.
- (A) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
(C) Statement-1 is True, Statement-2 is False
(D) Statement-1 is False, Statement-2 is True
4. Which of the following pairs of ions can't be present in the same solution ?
- (A) Al^{3+} and Cl^- (B) MnO_4^- and NO_2^-
(C) MnO_4^- and $\text{Cr}_2\text{O}_7^{2-}$ (D) SO_4^{2-} and Ni^{2+}
-

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

5. Which of the following will give colour with neutral FeCl_3 ?

- (A) HCOOH (B) CH_3COOH (C) PhCOOH (D) PhOH

6. Which of the following metal nitrate does NOT produce gaseous product on reaction with KCN ?

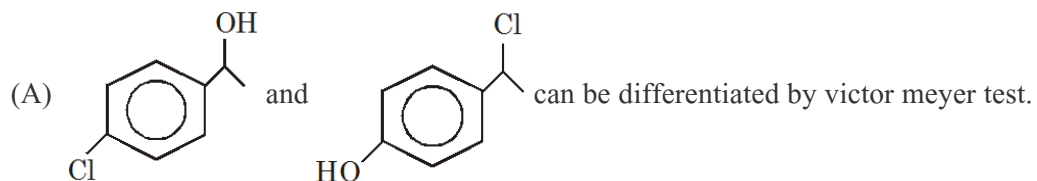
- (A) $\text{Cu}(\text{NO}_3)_2$ (B) AgNO_3 (C) $\text{Cd}(\text{NO}_3)_2$ (D) $\text{Pb}(\text{NO}_3)_2$

Space for Rough Work

7. Which of the following **does not** exist ?

- (A) TlI_3 $(+3)$ (B) CrI_3 $(+1)$ (C) FeI_3 $(+3)$ (D) BiF_5 $(+5)$

8. Select **correct** option.



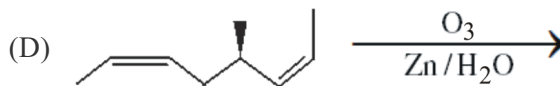
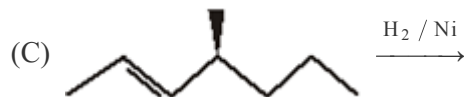
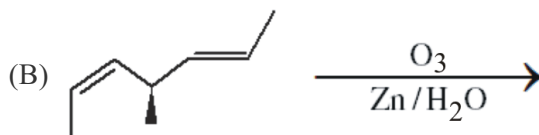
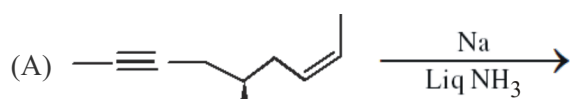
- (B) Monomer of orlon is acrylonitrile
 (C) D - glucose, D-fructose and L-Mannose give same osazone with $Ph-NH-NH_2$
 (D) Boiling point order of $CH_3 - OH > CH_3 - NH_2 > CH_4$

9. Which of the following is formed during partial roasting of $CuFeS_2$?

- (A) Cu_2S (B) FeO (C) SO_2 (D) $CuSO_4$

Space for Rough Work

10. In which of the following reaction the product will also be optically active



11. The resistance of 0.1 N solution of formic acid is 200 ohm and cell constant is 2.0 cm^{-1} , the equivalent conductance at infinite dilution is $1000 \text{ S cm}^2 \text{ eq}^{-1}$ of formic acid. Choose the **correct** statement(s).

- (A) Λ_{eq} of solution = $100 \text{ S cm}^2 \text{ eq}^{-1}$
 (B) degree of dissociation of electrolyte is 0.1
 (C) pH of solution is 2
 (D) specific conductance of solution is 10^{-2} Scm^{-1}

12. Which one of the following statement is **correct** ?

- (A) The net increase in entropy of a system is zero in any reversible Cyclic process.
 (B) For an adiabatic process entropy change of surrounding is zero.
 (C) The change in Gibb's free energy with pressure for one mole gas at constant temperature T is $\Delta G = RT \ln \frac{P_2}{P_1}$, where P_1 is initial pressure and P_2 is final pressure
 (D) For a spontaneous change $(\Delta G)_{T,P} > 0$

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

-
1. A mixture of A and B forms an ideal solution and has a vapour pressure of 200 mm of Hg when 33% of A is present. Another mixture with same components having 33% of B has a vapour pressure of 250 mm of Hg. Find vapour pressure of pure A in mm of Hg.
 2. Molecular formula weight of smallest ester which contains one chiral carbon is _____.
[Do not consider isotopes]
 3. How many minimum H-atoms are required to observe maximum number of distinct spectral lines when electron jumps from O shell to K shell?
-

Space for Rough Work

4. Number of planar species showing fractional bond order of either of X–O / O–O / X–X (X = central atom) is
 CO_3^{-2} , PO_4^{-3} , SO_4^{-2} , $\text{CH}_3\text{COO}^\ominus$, C_6H_6 , $\text{H}_2\text{PO}_2^\ominus$, O_3
5. Number of compounds in which orbitals used for overlapping by central atom is equal to orbitals used by central atom for hybridisation.
 PCl_5 , XeO_4 , H_2SO_4 , POCl_3 , IF_5 , XeO_6^{-4} , $\text{ClO}_3^\ominus$
6. A chalk sample exactly requires 200 ml of 7.3% (w/w) HCl solution of specific gravity 1.2, for complete reaction with all CaCO_3 present in it. If the chalk sample is 72% pure, the mass of sample taken (in g) is :
[Atomic mass : Ca = 40, Cl = 35.5]
-

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. If Z_1, Z_2, Z_3 are all unimodular and $Z_1 \cdot (\bar{Z}_2 + \bar{Z}_3)$ is purely imaginary, then $|Z_1 + Z_2 + Z_3|$ is
 (A) 1 (B) 2 (C) $\sqrt{3}$ (D) $\sqrt{6}$
2. In a right angled triangle, S_a and S_b denotes the medians to the legs of the triangle and the median belonging to the hypotenuse is S_c . The maximum value of the expression $\left(\frac{S_a + S_b}{S_c}\right)^2$ is
 (A) 8 (B) 4 (C) 10 (D) 5

Space for Rough Work

3. A series of hyperbolas is drawn by taking axes along reference axes having a common transverse axis of length $2a$ and centre at origin. If the locus of a point P on each hyperbola such that its distance from the transverse axis is equal to its distance from asymptote, is the curve $(x^2 - y^2)^2 = kx^2(x^2 - a^2)$, then the value of k is

(A) 4 (B) 3 (C) 2 (D) None of these

4. Let $f : \mathbb{R}^+ \rightarrow \mathbb{R}$ be a differentiable function $\forall x \in \mathbb{R}^+$. Where $f(x) = e + \ln\left(\frac{x}{e}\right)^{1-x} + \int_1^x f(t)dt \forall x \in \mathbb{R}^+$.

If the area enclosed by the curve $g(x) = x(f(x) - e^x)$ lying in the fourth quadrant is A , then A is equal to

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{1}{4}$ (D) $\frac{2}{3}$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

5. If the equation $x^2 + (2 - \tan\theta)x - (1 + \tan\theta) = 0$ has two integral roots, then possible sum of any two values of $\theta \in [0, 2\pi]$ is:

(A) π (B) 2π (C) $\frac{3\pi}{2}$ (D) $\frac{5\pi}{2}$

6. Let P and Q be two square matrix of order 3×3 such that $PQ = I - Q^2P^2$ and $P^3 = -Q^3$ then

(A) $QP + P^2Q^2 = I$ (B) $P - Q^2$ is non singular
 (C) $Q - P^2$ is non singular (D) $P - Q^2$ is singular

Space for Rough Work

7. A player tossed a coin. He sets one point for head and two points for tail. He plays till he gets of sum of points equal to n . If P_n be the probability that his score becomes 'n' then
- (A) $P_3 = \frac{1}{2}$ (B) $P_n = \frac{1}{2}P_{n-1} + \frac{1}{4}P_{n-2}$
- (C) $P_4 = \frac{11}{16}$ (D) $P_n = \frac{1}{2}(P_{n-1} + P_{n-2})$
8. Three different parabolas P_1 , P_2 and P_3 have same common tangent $x = 1$, $y = 2$ and $2x + y = 6$ in different regions and S_1, S_2, S_3 are respective focii of P_1, P_2 and P_3 . If (a, b) and r is centre and radius of a circle passing through the point S_1, S_2 and S_3 , then
- (A) $2a + b = 6$
- (B) $4r^2 = 5$
- (C) Max area of a triangle formed by the points S_1, S_2, S_3 is $\frac{15\sqrt{3}}{16}$
- (D) Max area of a triangle formed by the point S_1, S_2 and S_3 is $\frac{15\sqrt{3}}{4}$
9. Let $a, b, c > 0$ and $(a + b + c)^2 = 36$, then $\left(a + \frac{1}{b}\right)^2 + \left(b + \frac{1}{c}\right)^2 + \left(c + \frac{1}{a}\right)^2$ may be equal to :
- (A) $\frac{75}{4}$ (B) 35 (C) 15 (D) 10

Space for Rough Work

10. Let $f(x)$ is non-constant thrice differential function defined on $(-\infty, \infty)$ such that $f\left(\frac{x+15}{2}\right) = f\left(\frac{3-x}{2}\right)$ and $f'(0) = f'\left(\frac{3}{2}\right) = f'(3) = f'(5) = f'\left(\frac{11}{2}\right) = 0$. If the minimum number of zeros of $g(x) = f'(x)f'''(x) + f''(x) \cdot f''(x)$ in the interval $[0, 11]$ is $2k^2$, then
- (A) $\lim_{x \rightarrow |k|} [1 + \sin(x - |k|)]f'(x)^{\frac{k^2}{x-|k|}} = 1$
- (B) $f'(k^2) = f'(|k| + 1) = \text{sum of all possible values of } k$.
- (C) $f'(k^2 - |k|) = 0$
- (D) $f''(k^2 - 3|k|) = f''(2^{|k|} + 1)$
11. Solution of the differential equation $x^2 dy - 2xy dx = x^3 y^3 dx + x^4 y^2 dy$ is/are
- (A) $y = kx^2 e^{x^2 y^2 / 2}$
- (B) $y = ky^2 e^{x^2 y^2 / 2}$
- (C) $y = \frac{k}{x^2} e^{3x^2 y^2 / 2}$
- (D) $\ln\left(\frac{y}{x^2}\right) = \frac{(xy)^2}{4} + e^c$
12. If the normal to $\frac{x^2}{A^2} + \frac{y^2}{B^2} = 1$ at the ends of the chords $ax + by = 1$ and $cx + dy = 1$ be concurrent, then which of the following options is/are correct
- (A) $A^2 ae = 1$
- (B) $B^2 bd = -1$
- (C) $A^2 ae - B^2 bd = 0$
- (D) $A^2 ac + B^2 bd = 2$

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

-
1. If z is a complex number such that $|z| = 1$ and maximum value of $|z^4 + z^3 - 2z^2i + z + 1|$ is P , then the value of $[P]$ is ([.] GIF)
 2. Let $f(x) = 2x^3 - 3x^2 - x + \frac{3}{2}$ and $I = \int_{1/8}^{7/8} f(f(x)) \, dx$, then the value of $6I$ is equal to :
 3. Let 'n' be a five digit number and 'm' be four digit number from 'n' by deleting its middle digit, then the number of 'n' such that n/m is an integer
-

Space for Rough Work

4. The shortest distance between any two opposite edges of tetrahedron formed by planes $y + z = 0$, $x + z = 0$ and $x + y = 0$ and $x + y + z = \sqrt{96}$ is:
5. Let f be a differentiable function satisfies $\int_0^x (x - t + 1) f(t) dt - \int_0^x (4t^3 + 2t) dt = 0 \quad \forall x \geq 0$ then the value of $\frac{12}{\pi} \int_0^1 \frac{e^x dx}{\frac{d}{dx}(f(x) \cdot e^x) + 10e^x} =$
6. If $\sum_{r=0}^9 {}^{20}C_{2r} \cdot {}^{20}C_{2r+2} = \frac{a}{a+1} [{}^{39}C_{21} - {}^{19}C_{10}]$ and $xy = 4a$, then total number of ordered pairs (x, y) is (where $x, y \in \mathbb{I}$).

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work